

ABES ENGINEERING COLLEGE, GHAZIABAD

Session (2025-26)

Semester-I

Fundamentals of Linear Algebra & Statistics (25AS102)

Unit (4 & 5) Statistical Technique-I &II

Contents: Measures of central tendency, Skewness, Kurtosis, Curve Fitting, Method of least squares, fitting of straight lines, fitting of second-degree parabola, Exponential curves, Correlation and Rank correlation, Regression Analysis: Regression lines of y on x and x on y, regression coefficients, properties of regressions coefficients and non-linear regression.

Q. No	Question Description (1 Marks)	BLT																
1.	A cooperative bank has two branches employing 50 and 70 workers respectively. The average salaries paid by two respective branches are Rs. 360 and Rs. 390 per month. Calculate the mean of the salaries of all the employees.	K2																
2.	Find the median of 6, 8,9,10,11,12,13.	K2																
3.	Find out the mode of the following marks obtained by 15 students in a class. 4,6,5,7,9,8,10,4,7,6,5,8,7,7,9.	K2																
4.	Find the arithmetic mean of the following frequency distribution. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>f</td><td>5</td><td>9</td><td>12</td><td>17</td><td>14</td><td>10</td><td>6</td></tr></table>	x	1	2	3	4	5	6	7	f	5	9	12	17	14	10	6	K2
x	1	2	3	4	5	6	7											
f	5	9	12	17	14	10	6											
5.	In an asymmetrical distribution mean is 16 and the median is 20. Calculate the mode of the distribution.	K2																
6.	The first three central moments of a distribution are 0.15, -31. Find the moment of coefficient of Skewness.	K2																
7.	The first two moments of a distribution about the value '2' of the variable are 1, 16. Show that mean is 3, and variance is 15.	K2																
8.	The fourth central moment is 48. What must be its standard deviation in order that the distribution be mesokurtic.	K2																
9.	Write the normal equations to fit a curve $y = ax^2 + b$ by least square method.	K2																
10.	Write the formula of Karl Pearson correlation coefficient and write the range of correlation coefficient.	K2																
11.	If covariance between x and y variables is 10 and the variance of x and y are respectively 16 and 9. Find the coefficient of correlation.	K2																
12.	The lines of regression of y on x and x on y are respectively $y = x + 5$ and $16x - 9y = 94$, find the correlation coefficient.	K2																
13.	If the regression coefficient is 0.8 and 0.2, What will be the value of coefficient of Correlation?	K2																
14.	If the regression coefficient is 0.8 and 0.8, What will be the value of coefficient of Correlation?	K2																
15.	What is the relation between the regression coefficients and the coefficient of Correlation?	K2																
16.	Write the normal equations to fit a curve $y = \frac{c_0}{x} + c_1\sqrt{x}$	K2																
17.	Write the formula for rank correlation in the case of tied ranks.	K2																
18.	The first three moments of a distribution are 6, 25,-41. Find the moment of coefficient of Skewness.	K3																
19.	The first three moments of a distribution about the value '2' of the variable are 1, 16 and -40. Show that mean is 3, and variance is 15.	K4																

20.	Find the regression coefficient of y on x for the following: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>2</td><td>4</td><td>5</td><td>4</td><td>5</td></tr></table>	x	1	2	3	4	5	y	2	4	5	4	5	K3								
x	1	2	3	4	5																	
y	2	4	5	4	5																	
Q. No	Question Description (3 Marks)	BLT																				
1.	Calculate the first four central moments and also comment upon Skewness and Kurtosis from the following data: <table><tr><td>Class-interval</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td></tr><tr><td>Frequency</td><td>1</td><td>4</td><td>3</td><td>2</td></tr></table>	Class-interval	0-10	10-20	20-30	30-40	Frequency	1	4	3	2	K3										
Class-interval	0-10	10-20	20-30	30-40																		
Frequency	1	4	3	2																		
2.	Calculate the first four central moments about the mean, Skewness and Kurtosis for the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr></table>	x	0	1	2	3	4	5	6	7	8	f	1	8	28	56	70	56	28	8	1	K3
x	0	1	2	3	4	5	6	7	8													
f	1	8	28	56	70	56	28	8	1													
3.	Compute Skewness and Kurtosis, if the first four moments of a frequency distribution about the value 4 of the variable are 1,4,10 and 45.	K3																				
4.	The first four moments of a frequency distribution about the value 4 of the variable are -1.5, 17, -30 and 80. Find the moments $\mu_1, \mu_2, \mu_3, \mu_4$ about mean. Also find β_1 and β_2 .	K3																				
5.	The first four moments of a frequency distribution about the value 2 of the variable are 2, 20, 40 and 50 respectively. Comment upon the skewness and kurtosis of the distribution.	K3																				
6.	The first four moments of a frequency distribution about the value 5 of the variable are 1, 2.5, 5.5 and 16 respectively. Find the four central moments, moments about origin and coefficient of skewness.	K3																				
7.	Determine the Skewness and Kurtosis for the following data: <table><tr><td>Marks</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td></tr><tr><td>No. of students</td><td>18</td><td>20</td><td>30</td><td>22</td><td>10</td></tr></table>	Marks	10-20	20-30	30-40	40-50	50-60	No. of students	18	20	30	22	10									
Marks	10-20	20-30	30-40	40-50	50-60																	
No. of students	18	20	30	22	10																	
8.	Find the coefficient of correlation from the following points of observation (1,3),(2,2),(3,5),(4,4),(5,6).	K3																				
9.	A random sample of 5 college students is selected and their grades in Mathematics and Statistics are found to be: <table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Mathematics</td><td>85</td><td>60</td><td>73</td><td>40</td><td>90</td></tr><tr><td>Statistics</td><td>93</td><td>75</td><td>65</td><td>50</td><td>80</td></tr></table> Calculate rank correlation coefficient.		1	2	3	4	5	Mathematics	85	60	73	40	90	Statistics	93	75	65	50	80	K4		
	1	2	3	4	5																	
Mathematics	85	60	73	40	90																	
Statistics	93	75	65	50	80																	
10.	Calculate the coefficient of correlation for the following heights (in inches) of fathers (X) and their sons (Y): <table><tr><td>X</td><td>65</td><td>66</td><td>67</td><td>67</td><td>68</td><td>69</td><td>70</td><td>72</td></tr><tr><td>Y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table>	X	65	66	67	67	68	69	70	72	Y	67	68	65	68	72	72	69	71	K3		
X	65	66	67	67	68	69	70	72														
Y	67	68	65	68	72	72	69	71														
11.	Fit a parabolic curve of second degree to the following data: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1</td><td>1.8</td><td>1.3</td><td>2.5</td><td>6.3</td></tr></table>	x	0	1	2	3	4	y	1	1.8	1.3	2.5	6.3	K3								
x	0	1	2	3	4																	
y	1	1.8	1.3	2.5	6.3																	
12.	Use the method of least squares, find the curve $y = ab^x$ that best fits the following data: <table><tr><td>X</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Y</td><td>8.3</td><td>15.4</td><td>33.1</td><td>65.2</td><td>127.4</td></tr></table>	X	2	3	4	5	6	Y	8.3	15.4	33.1	65.2	127.4	K3								
X	2	3	4	5	6																	
Y	8.3	15.4	33.1	65.2	127.4																	

13.	Use method of least squares, fit the curve $y = ab^x$ to the following data:	K3																		
	<table><tr><td>x</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>144</td><td>172.8</td><td>207.4</td><td>248.8</td><td>298.5</td></tr></table>	x	2	3	4	5	6	y	144	172.8	207.4	248.8	298.5							
x	2	3	4	5	6															
y	144	172.8	207.4	248.8	298.5															
14.	Using method of least squares, fit the curve $y = ax^2 + bx$ to the following data:	K3																		
	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>y</td><td>1</td><td>1.2</td><td>1.8</td><td>2.5</td><td>3.6</td><td>4.7</td><td>6.6</td><td>9.1</td></tr></table>	x	1	2	3	4	5	6	7	8	y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1	
x	1	2	3	4	5	6	7	8												
y	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1												
15.	Find an exponential curve form $PV^y = k$ for the following data by the method of least squares:	K3																		
	<table><tr><td>V</td><td>50</td><td>100</td><td>150</td><td>200</td></tr><tr><td>P</td><td>135</td><td>48</td><td>26</td><td>17</td></tr></table>	V	50	100	150	200	P	135	48	26	17									
V	50	100	150	200																
P	135	48	26	17																
16.	Using method of least squares, fit the curve $y = c_0x + \frac{c_1}{\sqrt{x}}$ to the following data:	K3																		
	<table><tr><td>x</td><td>0.2</td><td>0.3</td><td>0.5</td><td>1</td><td>2</td></tr><tr><td>y</td><td>16</td><td>14</td><td>11</td><td>6</td><td>3</td></tr></table>	x	0.2	0.3	0.5	1	2	y	16	14	11	6	3							
x	0.2	0.3	0.5	1	2															
y	16	14	11	6	3															
17.	Using the least square fit a curve of the form $f(x) = a + bx + cx^2$ to the following data:	K3																		
	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$f(x)$</td><td>1</td><td>4</td><td>10</td><td>17</td><td>30</td></tr></table>	x	0	1	2	3	4	$f(x)$	1	4	10	17	30							
x	0	1	2	3	4															
$f(x)$	1	4	10	17	30															
18.	Examine least squares to fit a curve of the form $y = ae^{bx}$ to the following data:	K3																		
	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>1</td><td>1.2</td><td>1.8</td><td>2.5</td><td>3.6</td></tr></table>	x	1	2	3	4	5	y	1	1.2	1.8	2.5	3.6							
x	1	2	3	4	5															
y	1	1.2	1.8	2.5	3.6															
19.	If $4x - 5y + 33 = 0$ and $20x - 9y = 107$ are two lines of regressions. Find mean values of x and y , the correlation coefficient and the standard deviation of y if the variance of x is 9.	K3																		
20.	In a partial destroyed laboratory record of an analysis of correlation data, the following result only are legible: Variance $x = 9$ and Regression equation are $8x - 10y + 66 = 0, 40x - 18y = 214$. Calculate (a) the mean value of x and y (b) the standard deviation of y (c) Co-efficient of correlation between x and y .	K3																		
21.	Two lines of regression are given by $x + 52y - 5 = 0$ and $2x + 3y - 8 = 0$ and $\sigma_x^2 = 12$ Calculate (a) the mean value of x and y (b) Variance of y (c) Co-efficient of correlation between x and y .	K3																		
22.	The following table gives age (x) in years of cars and annual maintenance cost (y) in hundred rupees. <table><tr><td>x</td><td>1</td><td>3</td><td>5</td><td>7</td><td>9</td></tr><tr><td>y</td><td>15</td><td>18</td><td>21</td><td>23</td><td>22</td></tr></table> Calculate the maintenance cost for a 4-year-old car after finding regression equation.	x	1	3	5	7	9	y	15	18	21	23	22	K3						
x	1	3	5	7	9															
y	15	18	21	23	22															

23.	From the following data, determine the equations of line of regression of y on x and x on y :	K3																					
	<table><tr><td>x</td><td>6</td><td>2</td><td>10</td><td>4</td><td>8</td></tr><tr><td>y</td><td>9</td><td>11</td><td>5</td><td>8</td><td>7</td></tr></table>	x	6	2	10	4	8	y	9	11	5	8	7										
x	6	2	10	4	8																		
y	9	11	5	8	7																		
24.	Fit a parabolic curve of regression of y on x to the following data:	K3																					
	<table><tr><td>x</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td><td>3.5</td><td>4.0</td></tr><tr><td>y</td><td>1.1</td><td>1.3</td><td>1.6</td><td>2.0</td><td>2.7</td><td>3.4</td><td>4.1</td></tr></table>	x	1.0	1.5	2.0	2.5	3.0	3.5	4.0	y	1.1	1.3	1.6	2.0	2.7	3.4	4.1						
x	1.0	1.5	2.0	2.5	3.0	3.5	4.0																
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1																
25.	Find the multiple regression equation of X_1 on X_2 and X_3 from the data given below.	K3																					
	<table><tr><td>X_1</td><td>3</td><td>5</td><td>6</td><td>8</td><td>12</td><td>10</td></tr><tr><td>X_2</td><td>10</td><td>10</td><td>5</td><td>7</td><td>5</td><td>2</td></tr><tr><td>X_3</td><td>20</td><td>25</td><td>15</td><td>16</td><td>15</td><td>2</td></tr></table>	X_1	3	5	6	8	12	10	X_2	10	10	5	7	5	2	X_3	20	25	15	16	15	2	
X_1	3	5	6	8	12	10																	
X_2	10	10	5	7	5	2																	
X_3	20	25	15	16	15	2																	
26.	For the data given, determine the lines of regression:	K3																					
	<table><tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y</td><td>5</td><td>7</td><td>9</td><td>8</td><td>11</td></tr></table>	x	2	4	6	8	10	y	5	7	9	8	11										
x	2	4	6	8	10																		
y	5	7	9	8	11																		
27.	If $3x + 2y = 26$ and $6x + y = 31$ are two lines of regressions. Find (i) mean values of x and y (ii) the coefficient of correlation between x and y (iii) find variance of y if the variance of x is 9.	K5																					
28.	Fit the curve $y = ae^{bx}$.	K4																					
	<table><tr><td>x</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y</td><td>4.077</td><td>11.084</td><td>30.128</td><td>81.897</td><td>222.62</td></tr></table>	x	2	4	6	8	10	y	4.077	11.084	30.128	81.897	222.62										
x	2	4	6	8	10																		
y	4.077	11.084	30.128	81.897	222.62																		
29.	Calculate all four moments about mean and also Skewness and Kurtosis.	K4																					
	<table><tr><td>Marks</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td></tr><tr><td>No. of students</td><td>1</td><td>6</td><td>10</td><td>15</td><td>11</td><td>7</td><td>10</td></tr></table>	Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	No. of students	1	6	10	15	11	7	10						
Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70																
No. of students	1	6	10	15	11	7	10																
30.	The first four moments of a frequency distribution about the value 28.5 of the variable are 0.294, 7.144, 42.409 and 454.98 respectively. Calculate the moments about mean. Also evaluate β_1 and β_2 and comment upon the skewness and kurtosis of the distribution.	K3																					
31.	For 10 observations on price (x) and (y), the following data were obtained (in appropriate units): $\sum x = 130, \sum y = 220, \sum x^2 = 2288, \sum y^2 = 5506$ and $\sum xy = 3467$. Obtain the two lines of regression and estimate the supply when the price is 16 units.	K3																					
32.	The velocity V of a liquid is known to vary with temperature T , according to a quadratic law $V = a + bT + cT^2$. Find the best values of a, b, c for the following data:	K3																					
	<table><tr><td>T</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>V</td><td>2.31</td><td>2.01</td><td>1.80</td><td>1.66</td><td>1.55</td><td>1.47</td><td>1.41</td></tr></table>	T	1	2	3	4	5	6	7	V	2.31	2.01	1.80	1.66	1.55	1.47	1.41						
T	1	2	3	4	5	6	7																
V	2.31	2.01	1.80	1.66	1.55	1.47	1.41																
33.	The following results were obtained from records of age (x) and systolic blood pressure (y) of a group of 10 men: and $\sum(x - \bar{x})(y - \bar{y}) = 1120$. Find the appropriate regression and use it to estimate the blood pressure of a man whose age is 45.																						
	<table><tr><td></td><td>x</td><td>y</td></tr><tr><td>Mean</td><td>53</td><td>142</td></tr><tr><td>Variance</td><td>130</td><td>165</td></tr></table>		x	y	Mean	53	142	Variance	130	165													
	x	y																					
Mean	53	142																					
Variance	130	165																					